07 Metrics Suite - Full Evaluation Metric Suite

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0.1 Introduction

This notebook applies a full metric suite following Wang et al. (2025) and Van Calster et al. (2019).

Primary: PhysioNet Utility Score (Reyna et al., 2020), calibration (intercept + slope).

Secondary: AUPRC, AUROC, decision-curve net benefit, alert burden (≤ 1.4 benchmark; Moor et al., 2023), median lead time.

All metrics are computed in two windows: **unrestricted** (all test hours) and **treatment-anchored** (pre-treatment hours only; Kamran et al., 2024).

0.2 Metric Functions

Standalone metric functions, reused across internal, external, and subgroup evaluations. Keeping one implementation matters more than it looks: internal and external AUROC are differenced in H5, and per-hour and per-patient operating points are compared in stage 11. Two scoring paths that disagreed even slightly would put that disagreement into the contrast.

0.3 Evaluation Loop

The full models x variants x windows grid is evaluated. Decision-curve analysis runs where both primary and GBT predictions are available.

0.4 Results and Interpretation

The central performance comparison. Three readings are worth making from the table, and each answers a different question.

Primary against GBT on the Utility Score speaks to H6, though the two models can order differently on utility than on AUROC; where they do, the disagreement is itself informative and the thesis reports both orderings rather than choosing one.

Treatment-anchored against unrestricted speaks to H3. A gap here would be leakage from clinician suspicion. No gap is harder to interpret than it looks, because under a conjunction onset rule the pre-treatment window may hold no events at all, in which case the comparison has nothing to measure.

Calibration slope is read against 1.0. A slope well below 1 means predicted risks are too extreme at both ends, which a discrimination metric cannot see and which matters for any threshold a clinician would actually deploy.

```
##
## Saved: 07_metric_results.parquet
```

0.5 Utility Score Across the Threshold Grid

Every Utility Score in the table above is read at the default 0.5 cut-off. At an hourly event rate below 0.5 per cent no model's predicted probability comes near 0.5, so that cut-off measures the no-alert strategy for some models and an arbitrary point for the others. The full grid is written to `07_utility_sweep.csv`; the maximum over all thresholds that raise at least one alert is the honest statement of what each model can achieve.

Two invariants are checked here rather than asserted: a threshold above every predicted probability must score exactly 0 (it *is* the no-alert strategy), and the normalised score must never exceed 1.

```
## Check: 10 no-alert operating points all score exactly 0.
## Check: no operating point exceeds the optimal-strategy bound of 1.
##
##
## Saved: 07_utility_sweep.csv, 07_utility_max.csv
```

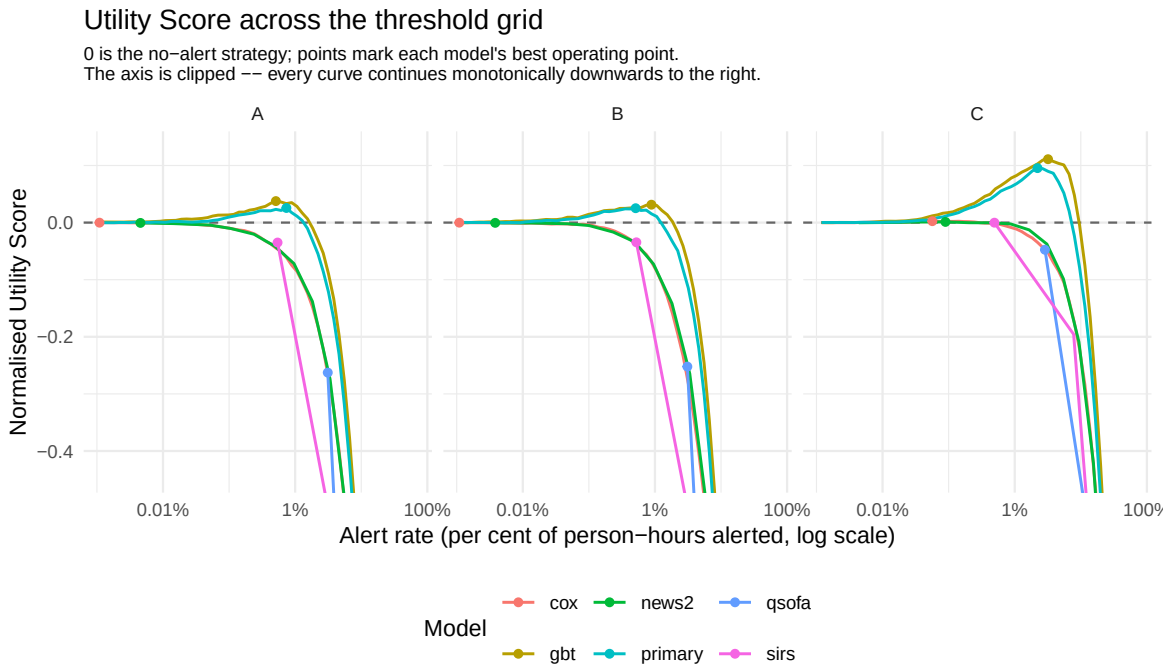

Figure 1: Normalised Utility Score against alert rate for every model and pre-registered label variant on the temporal test set. The dashed line at zero is the no-alert strategy: a curve below it describes a configuration that would be better switched off. Points mark each model's best operating point. The vertical axis is clipped, and every curve continues monotonically downwards to the right.

0.6 Alternative-Label Evaluation (Protocol Amendment 2)

Variants D and E are evaluated with the identical metric suite. Because a label change alters the event definition, the event counts and prevalences differ from the Sepsis-3 variants and are reported alongside the metrics; the comparison of interest is made in `09_variance_decomposition.Rmd`.

```
##
## Saved: 07_metric_results_altlabel.parquet
```

0.7 Feature-Ablation Evaluation (Protocol Amendment 5)

The ablated fits from stages 05 and 06 are scored with the identical metric suite on the identical test rows, so the difference between the two arms is attributable to the feature set alone. The ablation removes the action-derived features — vasopressor exposure and the per-test order indicators — and retains all physiological measurements, including forward-filled laboratory values.

```
##
## Saved: 07_ablation_metrics.parquet / .csv
```

0.8 Random-Split Evaluation (H4: Split Optimism)

The same metric suite is applied to random-split predictions (Variant B only). The output `07_metric_results_random.parquet` feeds the variance decomposition for H4 (Guo et al., 2022).

```
##
## Saved: 07_metric_results_random.parquet
```